DSF182: Data Science in Finance

Summative Submission 3 Part B

Case Study Analysis

Student name	Gopolang Mmutlwane
Module	DSF182 – Data Science in Finance
Assessment	SS3, Part B
Case study	Wang, J. and Gu, M. (2026) 'Optimizing risk profiling of agricultural loans: default prediction using multi-source remote sensing digital footprints'
AI conversation link	https://claude.ai/share/61e93de4-793b-4396-a256-ddcdc45c32cb

Case study: Optimizing Risk Profiling of Agricultural Loans: Default Prediction Using Multi-Source Remote Sensing Digital Footprints (Wang and Gu, 2026)

DOI: https://doi.org/10.3390/ijfs14070187

Question 1: Do you note any issues in the case study that could compromise reproducibility?

- **Confidential loan data.** The loans come from an unnamed "land management platform in Chongqing Municipality" (Wang and Gu, 2026, p. 7). They are "not publicly available due to privacy and project confidentiality restrictions" and are provided only "upon reasonable academic request" (Wang and Gu, 2026, p. 37).
    - *Inference:* the pipeline cannot be rerun on the original loans without the authors' approval.
- **A manual geocoding step.** Failed addresses were "manually supplemented by cross-checking the original loan file location descriptions with online maps" (Wang and Gu, 2026, p. 15). I found no rule for this step, and no statement on whether these addresses passed the 95% confidence filter (Wang and Gu, 2026, p. 19, Table 8).
    - *Inference:* another researcher could place some loans differently and attach different climate values to them.
- **Missing code and settings.**
    - I found no code or repository link; the Data Availability Statement covers data only (Wang and Gu, 2026, p. 37).
    - Versions are given only for Python 3.13.9 and the R package SPEI 1.8.1 (Wang and Gu, 2026, p. 16); I found none for the modelling libraries.
    - Table A5 gives hyperparameter search ranges (Wang and Gu, 2026, p. 40), but I found no final values.
- **An unclear target variable.**

- The broad default definition includes “Special Mention” loans (Wang and Gu, 2026, p. 9), but the same 418 defaults are later described as “‘substandard’ or below… or already written off” (Wang and Gu, 2026, p. 18).
- The narrow definition starts at “Substandard” (Wang and Gu, 2026, p. 9), but later at “‘doubtful’ or below” (Wang and Gu, 2026, p. 23).
- Land-rights collateral is 1.97% of loans with a 24.58% default rate in the text (Wang and Gu, 2026, p. 10), but 266 loans (2.93%) with a 21.80% default rate in Table 3 (Wang and Gu, 2026, p. 8).
- *Inference:* a replicator cannot be sure which default label the models were trained on, and every reported score depends on it.

Question 2: What decision or process is being improved, and for whom?

- **The process being improved:**
  - Traditional risk control “relies on borrowers’ static financial statements, limited credit histories, and traditional collateral valuation”, making it hard for institutions “to price risks precisely” (Wang and Gu, 2026, p. 2).
  - It also has “a limited capacity to capture exogenous variables such as climate risk”, so the study adds “multi-source remote sensing data with explainable machine learning” (Wang and Gu, 2026, p. 1).
- **The decisions supported:** “more informed pre-loan screening and dynamic in-loan monitoring” (Wang and Gu, 2026, p. 3).
  - *Inference:* in practice, pre-loan screening means deciding whether, and on what terms, to lend to a farmer.
- **For whom:**
  - *Financial institutions* are the direct users; the authors offer them “a reusable feature system and model framework” (Wang and Gu, 2026, p. 3).
  - *Regulators:* the framework could help build a “regional agricultural financial risk monitoring platform” supporting “the monitoring of systemic risk at the macro level” (Wang and Gu, 2026, pp. 3–4). The authors did not build one.

- *Inference:* here the decision-maker is the regulator, and the decision is regional supervision rather than loan approval.

- *Farmers (inference):* conservative lending "is associated with raising credit access thresholds and shortening loan maturities" (Wang and Gu, 2026, p. 4), so better risk profiles could widen farmers' access to credit, in line with the paper's aim of "balancing financial inclusion and safety" (Wang and Gu, 2026, p. 1).

Question 3: With respect to the data that the study uses, what do the records represent, what is missing and what may be sensitive?

- **What the records represent:**
  - One row is one agricultural loan contract, labelled default or non-default using the regulator's (CBIRC) five-tier loan classification (Wang and Gu, 2026, p. 9).
  - There are 9,070 contracts from 8,221 operators: "individual farmers, family farms, and farmer-specialized cooperatives" (Wang and Gu, 2026, p. 8), covering 2016–2018 and "all 37 districts and counties of Chongqing" (Wang and Gu, 2026, p. 7).
  - Each row holds borrower, loan and collateral variables (Wang and Gu, 2026, p. 7, Table 1). Defaults number 418 (4.61%) under the broad definition and 237 (2.61%) under the narrow one (Wang and Gu, 2026, p. 9).
  - Satellite and climate features come from the calendar year before origination (t − 1), with climate taken over its March–October growing season (Wang and Gu, 2026, p. 17).
- **What is missing:**
  - *Parcel boundaries:* there is no "parcel boundary information in the original agricultural loan data" (Wang and Gu, 2026, p. 15).
  - *Excluded loans:* screening removed 686 of 9,756 contracts, including 215 that could not be located (Wang and Gu, 2026, p. 19, Table 8).
  - *One province only:* the sample "is concentrated in a single province" (Wang and Gu, 2026, p. 35).

- *Inference:* as described in Table 1, the dataset has no repayment history, age or education variables (Wang and Gu, 2026, p. 7), so the model does not use these traditional factors.

- **What may be sensitive:**
  - Household income, loan amounts and default status (Wang and Gu, 2026, p. 8, Tables 2–3).
  - Collateral addresses “typically recorded at the village level” (Wang and Gu, 2026, p. 14), which were sent to the Baidu Map geocoding service (Wang and Gu, 2026, p. 15).
    - *Inference:* borrower addresses were shared with a third party, and coordinates combined with income could identify a household.
  - Ethics review and consent are both “Not applicable” (Wang and Gu, 2026, p. 37).

Question 4: Which method is being used for modelling and why is this method suited to the pattern or decision?

- **A three-stage design:** “association testing—predictive comparison—interpretation of feature contributions” (Wang and Gu, 2026, p. 19).
- **Stage 1, logistic regression:** “suitable for binary dependent variables” (Wang and Gu, 2026, p. 20), it tests each variable’s link to default “rather than pursuing predictive accuracy” (Wang and Gu, 2026, p. 19).
- **Stage 2, eight machine learning models:**
  - A decision tree, a random forest, four boosting models (GBM, XGBoost, LightGBM, CatBoost) and two neural networks (MLP, TabNet). Boosting works “by iteratively fitting residuals” (in plain terms, each new tree corrects earlier errors) and is “adept at capturing complex nonlinear relationships and interaction patterns among features” (Wang and Gu, 2026, p. 21).
  - A pool is used so that conclusions hold “under different modeling approaches” (Wang and Gu, 2026, p. 21).

    - Random forest scores the highest ROC-AUC, but LightGBM is “selected as the representative model” (Wang and Gu, 2026, pp. 30–31).
- **Stage 3, the Explainable Boosting Machine (EBM):** the authors choose it because, they say, its accuracy is “comparable to that of complex models such as random forest and gradient boosting” (Wang and Gu, 2026, p. 24). I found no EBM performance score in the paper.
- **Why it suits the pattern and the decision:**
    - *Rare defaults:* the authors respond to the 4.61% default rate with class weights and a threshold that maximises Youden’s J (Wang and Gu, 2026, p. 22).
    - *Auditable decisions:* the EBM expresses predictions “as the sum of individual feature contributions”, “meeting financial regulatory requirements for model auditability and interpretability” (Wang and Gu, 2026, p. 24).
    - *Inference:* climate effects on repayment are unlikely to be linear, which suits the nonlinear models, and a lender must be able to explain a decline, which the EBM’s additive form allows.

Question 5: How was the model’s performance evaluated, specifically what baseline and measures were used?

- **Test design:** a 7:3 split grouped by farmer ID, so no farmer appears in both sets (Wang and Gu, 2026, p. 22); the test set “is used only once during the final evaluation” (Wang and Gu, 2026, p. 21).
- **Baselines:**
    - *Data baseline:* Dataset A (traditional features only) is “the reference benchmark without remote sensing information”, compared with Datasets B and C, which add satellite and climate features (Wang and Gu, 2026, p. 21).
    - *Model baseline:* the decision tree serves “as the baseline reference for performance comparison” (Wang and Gu, 2026, p. 21).
- **Measures, and why they were chosen:**

    - ROC-AUC is the principal criterion because it “is insensitive to class imbalance”; PR-AUC goes to the appendix because its baseline “equals the positive rate (4.61% in this dataset)” (Wang and Gu, 2026, p. 22).
    - The Brier score measures calibration and the KS statistic discrimination (Wang and Gu, 2026, p. 23).
    - Because “the default equal-probability threshold often misclassifies most defaulted loans as normal” (Wang and Gu, 2026, p. 22), sensitivity, specificity and accuracy are reported at a Youden’s J threshold (Wang and Gu, 2026, p. 23).
- **Results:** Dataset A models scored an AUC of 0.66–0.75 (Wang and Gu, 2026, p. 29). Adding all remote sensing features raised AUC by “approximately 0.11”; random forest rose from 0.733 to 0.865 (Wang and Gu, 2026, p. 28, Table 11), with a sensitivity of 0.776 and specificity of 0.797 (Wang and Gu, 2026, p. 30, Table 12).
    - *Inference:* the model catches about 78% of defaults but also flags about 20% of good loans.
- **Gaps:**
    - The authors state they used a bootstrap for AUC confidence intervals (Wang and Gu, 2026, p. 23), but I found no results, though the top three models differ only “in the order of thousandths” (Wang and Gu, 2026, p. 30).
    - *Inference:* logistic regression, a common industry benchmark, is absent from the predictive comparison (Wang and Gu, 2026, p. 28, Table 11), and the split is by farmer, not by time.

Question 6: With respect to using the model in the real world, how would the model’s output become an action?

- **Proposed, not deployed.** I found no description of a lender using the model; model choice “for practical deployment should be calibrated to risk preferences and business objectives” (Wang and Gu, 2026, p. 31).

- **From score to flag:** a loan is flagged when its predicted probability exceeds “the global threshold”, the mean of the Youden’s J thresholds across folds (Wang and Gu, 2026, p. 22).
- **Available at decision time:** t − 1 satellite data are public when a loan is made, “meeting the basic data availability constraints for pre-loan risk assessment” (Wang and Gu, 2026, p. 17).
- **Recommended actions:**
    - *Pre-loan:* climate variability used “as reference factors in customer screening and eligibility review” (Wang and Gu, 2026, p. 36).
    - *Loan terms:* for “high climate volatility with relatively insufficient collateral coverage”, “risk premiums may be moderately raised or lower loan-to-value ratios required” (Wang and Gu, 2026, p. 36).
    - *Post-loan:* “early-warning mechanisms can be established based on risk thresholds identified by interpretable models”; a full “dynamic early-warning module” is left as future work (Wang and Gu, 2026, p. 36).
    - *Inference:* these act on the climate indicators and EBM-derived thresholds, not directly on the predicted default probability.
- **What a flag means:** on the test set, LightGBM detected 97 defaults but produced 537 false positives (Wang and Gu, 2026, p. 43).
    - *Inference:* only about 15% of flagged loans (97 of 634) defaulted, so in my view a flag should prompt human review or adjusted terms rather than an automatic decline.
- **Regulators:** the authors recommend disclosure of “the alternative data and modeling methods” (Wang and Gu, 2026, p. 36) and auditing standards covering “model stability, fairness, and interpretability” (Wang and Gu, 2026, p. 37).

Question 7: If the model they developed would fail, who may be affected and how would the problem be detected?

- **How the model could fail:**

- *Sample and data limits:* the sample is from “a single province”, satellite accuracy is limited by “observational errors”, and point matching “may introduce spatial correspondence bias” (Wang and Gu, 2026, p. 35).
- *Unexpected patterns:* loans with drought stress defaulted less often (2.52%) than those without (5.59%), “contrary to the expectation” (Wang and Gu, 2026, p. 12).
    - *Inference:* the model may have learned a 2016–2018 pattern that fails when the climate or lending practice changes.

- **Who may be affected** (LightGBM, the representative model (Wang and Gu, 2026, p. 31), on the test sample):
    - *Farmers wrongly flagged:* 537 non-defaulting loans were flagged (Wang and Gu, 2026, p. 43) and could face higher “risk premiums” or “lower loan-to-value ratios” (Wang and Gu, 2026, p. 36). I found no discussion of this cost.
        - *Inference:* creditworthy farmers could pay more or borrow less, the “credit rationing” the paper aims to reduce (Wang and Gu, 2026, p. 2).
    - *Lenders:* sensitivity is 0.776 (Wang and Gu, 2026, p. 30, Table 12).
        - *Inference:* about 22% of test-sample defaults went unflagged, which in real use means losses.
    - *The financial system:* the authors claim the framework could support “the monitoring of systemic risk at the macro level” (Wang and Gu, 2026, p. 4).
        - *Inference:* if regulators relied on a failing model, region-wide climate shocks could be underestimated.
- **How the problem would be detected:**
    - *Auditing (a precondition, not an alert):* the EBM’s “auditability and interpretability” (Wang and Gu, 2026, p. 24) allow inspection but do not themselves detect failure.
    - *Proposed checks:* “verification and auditing standards” (Wang and Gu, 2026, pp. 36–37), and testing “across different agricultural ecological zones” (Wang and Gu, 2026, p. 36).

- I found no plan for monitoring the model's own accuracy once in use.
    - *Inference:* a lender could compare predicted with actual defaults each season, but defaulted loans had contract terms of about 637 days (Wang and Gu, 2026, p. 9) and I found no time-to-default figure, so a failing model might go unnoticed for months.

Reference list

Wang, J. and Gu, M. (2026) 'Optimizing risk profiling of agricultural loans: default prediction using multi-source remote sensing digital footprints', *International Journal of Financial Studies*, 14(7), 187. Available at: https://doi.org/10.3390/ijfs14070187 (Accessed: 13 October 2026).